

# Andrew Peterson

QUANTUM RESEARCHER · SECURITY ENGINEER · SOFTWARE ARCHITECT

California, United States of America

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## Summary

I'm a full-stack software engineer who is driven by the overlap between emerging technology like quantum computing and artificial intelligence, and reliable, secure infrastructure. Over the past few years, I've focused on bridging the gap between the bleeding edge and production- and consumer-ready code, whether that be architecting quantum-encrypted video platforms or building cloud-native backends for global cybersecurity competitions. I enjoy the challenge of high-consequence systems and have a track record of founding organizations and leading teams to make this emerging tech more accessible. I'm looking to bring that mix of curiosity and engineering rigor to a team solving genuinely difficult problems that can change the world.

## Education

### University of California: San Diego

B.S. IN COMPUTER SCIENCE

La Jolla, California

Sep. 2020 - Dec. 2024

## Skills

|                              |                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>             | Python (Expert), C/C++, Java, TypeScript/JavaScript, SQL, Bash, LaTeX, Matlab, Rust                           |
| <b>Quantum Computing</b>     | Qiskit, Cirq, Quantum Neural Networks (QML), QAOA, Grover's & HHL Algorithms, BB84, Error Mitigation          |
| <b>Systems &amp; DevOps</b>  | Docker, Kubernetes, Terraform, Linux Kernel, Distributed Systems, CI/CD, Nginx, Git, AWS, DigitalOcean        |
| <b>Cybersecurity</b>         | Reverse Engineering (Ghidra/IDA), Network Security (TCP/IP), Penetration Testing, OSINT, Applied Cryptography |
| <b>Software Engineering</b>  | Data Structures & Algorithms (DSA), System Design, Design Patterns, Agile/Scrum, Unit Testing                 |
| <b>Data Science &amp; AI</b> | PyTorch, Scikit-Learn, Pandas, Computer Vision, NLP, Ollama (LLMs), Statistical Modeling, Jupyter             |
| <b>Infrastructure</b>        | PostgreSQL, MongoDB, Redis, RESTful/GraphQL API Design, Microservices, Linux Server Administration            |
| <b>Experimental Lab</b>      | Free-space Photonics, Laser Systems, Signal Processing, Oscilloscopes, Atomic Force Microscopy (AFM)          |

## Experience

### Qualcomm Institute

QUANTUM SOFTWARE ENGINEERING LEAD (INTERN)

La Jolla, California

Jul. 2022 - Dec. 2024

- Architected and spearheaded a video communication platform utilizing end-to-end quantum encryption via photonic circuitry, enhancing data security protocols.
- Optimized Grover's algorithm using Qiskit to solve complex binary constraint satisfaction problems, demonstrating practical applications for quantum search.
- Developed and evaluated Quantum Machine Learning (QML) models, implementing classification algorithms via quantum neural networks to improve predictive accuracy.
- Mentored a cross-functional team of interns under Dr. Nikola Alic, overseeing software deliverables and the technical setup of a new undergraduate quantum computing lab.

### Rochester Institute of Technology Esports

FULL-STACK DEVOPS ENGINEER & ADVISOR

Remote (Rochester, New York)

Sep. 2020 - Jul. 2022

- Engineered and deployed scalable web resources using the MERN stack (MongoDB, Express, React, Node.js), leveraging Docker for consistent environment orchestration.
- Orchestrated cloud infrastructure on DigitalOcean Droplets, implementing Nginx reverse proxies and Docker Compose to ensure 99.9% uptime for high-traffic services.
- Streamlined development lifecycles by building CI/CD pipelines and monitoring suites, reducing deployment friction and enhancing system performance.

### Mathnasium

INSTRUCTOR & IT SYSTEMS SPECIALIST

Temecula, California

Mar. 2020 - Sep. 2021

- Delivered advanced technical instruction in Calculus and Computer Science, tailoring complex concepts to diverse student learning levels.
- Directed the center's digital transformation during COVID-19, authoring technical documentation and maintaining IT infrastructure to enable seamless remote learning.

# Extracurricular Activity

## Quantum Computing at UC San Diego

La Jolla, CA

Co-FOUNDER & PRESIDENT (2023 - 2024)& TECHNICAL CURRICULUM LEAD

Feb. 2021 - Dec. 2024

- Co-founded and scaled the first undergraduate quantum computing organization at UC San Diego, establishing a community for 100+ aspiring researchers and engineers.
- Developed a comprehensive technical curriculum for workshops covering foundations in quantum computing and Qiskit.
- Standardized organizational workflows and digital infrastructure, enabling sustainable leadership transitions and cross-campus collaborations.
- Presented components of internship to organization members, including quantum-encrypted communication and quantum search algorithms.

## ACM Cyber @ UCSD

La Jolla, California

PRESIDENT & CYBERSECURITY COMPETITION MEMBER

Aug 2020 - June 2023

- Trained teams for (and competed in) high-stakes national and international competitions including CCDC, CPTC, and Google CTF.
- Authored pedagogical training modules on penetration testing, network defense, and cryptography to elevate the club's competitive ranking.
- Orchestrated weekly technical workshops and "Deep Dive" sessions, fostering a culture of peer-to-peer mentorship in advanced security topics.

## San Diego Capture the Flag (SDCTF)

Remote

LEAD INFRASTRUCTURE ENGINEER & CHALLENGE DESIGNER

Dec. 2020 - May 2022

- Engineered the backend for San Diego's largest annual cybersecurity competition, supporting 500+ global participants in its inaugural year.
- Provisioned cloud-native infrastructure using Terraform (IaC), managing a Postgres database and containerized challenge environments.
- Designed and deployed a Discord.JS-integrated submission bot to automate real-time flag validation and leaderboard updates.
- Developed high-quality Open Source Intelligence (OSINT) challenges, testing competitors' ability to perform investigative digital forensics.

# Certifications

2025    **Azure Data Fundamentals**, Microsoft

2022    **IT Fundamentals+**, CompTIA

# References

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